

9.8 Notification System

The notification system provides real-time updates through an in-app notification center:

- A bell icon in the navigation header displays the unread notification count as a badge
- Clicking the bell opens a dropdown panel showing recent notifications
- Notification types include:
 - *Info*: General platform updates and announcements
 - *Success*: Successful operations (listing created, payment received)
 - *Warning*: Action required (new listing claimed, pickup started)
 - *Error*: Failed operations or system issues
- Each notification can be marked as read individually
- Notifications are sorted by creation time (newest first)

9.9 Carbon Credits Display

The carbon credits section provides environmental impact quantification:

- A dedicated panel on the farmer dashboard shows total carbon offsets earned
- Individual listing contributions are itemized with waste type, quantity, and calculated CO₂ savings
- A cumulative progress bar shows progress toward environmental milestones
- Carbon offset values are calculated using the scientifically established factors defined in the `CARBON_FACTORS` constant

9.10 Responsive Design

The application adapts its layout across three primary breakpoints:

- **Desktop (1024px+)**: Full-width layout with sidebar navigation, multi-column card grids, and side-by-side form layouts
- **Tablet (768px–1023px)**: Reduced column count (2-column grids), collapsible navigation, and stacked form fields
- **Mobile (below 768px)**: Single-column layout, hamburger menu navigation, full-width cards, and bottom sheet dialogs instead of modal dialogs

Tailwind CSS responsive modifiers (`sm:`, `md:`, `lg:`, `xl:`) are used consistently throughout the codebase to implement these breakpoints.

The figures above are referenced in the final output chapter and can be updated anytime by replacing image files in the report image folder.